



Roll No:

**MCA-INT
(SEM V) THEORY EXAMINATION 2024-25
CYBER SECURITY**

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data, choose suitably.

SECTION A

1. Attempt all questions in brief.

$$2 \times 07 = 14$$

Q no.	Question
a.	Explain the concept of Management Information Systems (MIS).
b.	How does Cyber Security differ from Information Security?
c.	How does a VPN improve network security?
d.	Explain the difference between a Worm and a Virus.
e.	Why is physical security important for IT assets?
f.	Discuss CCTV systems contribute to physical security?
g.	What are the key components of an Email Security Policy?

SECTION B

2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question
a.	What is Information Assurance? Discuss how it ensures the integrity, availability, and confidentiality of information in an organization.
b.	Describe how Digital Signatures work and their importance in secure online transactions. How do they ensure data integrity and authenticity?
c.	How do CCTV and Intrusion Detection Systems (IDS) contribute to overall security in an organization? Explain the benefits of combining these physical security measures with cybersecurity protocols.
d.	What are the key provisions of the Intellectual Property Rights (IPR) laws in relation to Information Security? Discuss how Copyright, Software Licenses, and Patents help protect intellectual property in the digital age.
e.	Discuss the role of Malware, including viruses, worms, and Trojan horses, in compromising Information Systems. What strategies can be employed to detect and eliminate these threats?

SECTION C

3. Attempt any *one* part of the following:

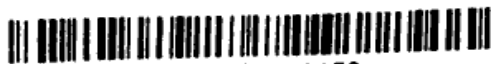
$$07 \times 1 = 07$$

Q no.	Question
a.	Discuss the process of conducting a Security Risk Analysis. What steps are involved in identifying, assessing, and mitigating security risks?
b.	Compare and contrast the concepts of Physical Security and Cyber Security. How do they work together to protect an organization's information resources?

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	Explain the concept and functioning of Cryptography. Explain its types?
b.	How do Firewalls and VPNs work together to secure remote connections and networks? What challenges do organizations face when implementing these security technologies?



PAPER ID-310150

Printed Page: 2 of 2
Subject Code: RCAI505

Roll No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

MCA-INT
(SEM V) THEORY EXAMINATION 2024-25
CYBER SECURITY

TIME: 3 HRS

M.MARKS: 70

5. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question
a.	Explain the concept of Security in Data Storage and the challenges it poses in modern information systems. Discuss how encryption, access control, and data masking can be used to protect stored data.
b.	Explain the importance of Risk Management in Information Security. How can organizations identify, assess, and mitigate risks to their information systems? Provide examples of risk management frameworks used in cybersecurity.

6. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question
a.	How do Cyber Laws in India address issues like cybercrime, data breaches, and unauthorized access? Discuss the role of the IT Act 2000 in curbing cybercrime and securing digital assets in India.
b.	How does software piracy impact information security and intellectual property protection? Discuss the legal and business consequences of software piracy and the role of organizations in preventing it.

7. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question
a.	Provide an overview of Corporate Security Policies. How do these policies ensure consistency and compliance across the organization? Discuss how to align corporate security policies with overall business goals and legal requirements.
b.	How does multi-factor authentication (MFA) enhance access control systems? Discuss its effectiveness in mitigating unauthorized access risks.